Exercise: Explain how the three proposed models on the previous slides seem to perform, in regards to the appropriate amount of smoothing of the histogram and scatter plot.

In each case, the dashed (blue) line is the smoothest model. In fact, in both cases the model is oversmoothing, masking seemingly important features in the data

The dash-dotted line (red) is the least smooth. It is fitting to fluctuations that are likely "artifacts"

The solid (black) line is a compromise between these two extremes

Exercise: Explain why overfitting is such a significant concern.

"overfitting" ⟺ "undersmoothing"

A key objective is to extend from the available sample ("training data") to the larger population from which it was drawn

An overfit model is fit to "artifacts" that are only present in the training sample the result will be predictions (forecasts) for the larger population which will have large errors

In many situations, there are one or more unknown population quantities that are of interest to estimate. For example,

- The probability that the return exceeds c tomorrow
- The "beta" for some stock over the past year.
- The volatility for some stock over the past three months.
- The mean of some distribution of interest.
- The variance of some distribution of interest.
- The λ parameter from some Exponential distribution.

The model serves as a means to estimating such a quantity, since we imagine that the true model, if known, would provide the true value of the unknown. An estimated model provides an estimate of the unknown.

From a statistical standpoint, we say that a probability model is incompletely specified. I.e., a particular model is assumed, but there are still unknown components, often real-valued parameters.

We use available data (a sample) to estimate these parameters.

The quantities of interest such as those listed above are either the parameters themselves, or some function of those parameters.

Example: In a certain situation, an individual is comfortable assuming that $X_1, X_2, \ldots X_n$ are independent and identically distributed (i.i.d.) with the Normal distribution with mean μ and variance σ^2. These parameters are unknown, and must be estimated from data.

Exercise: Suppose someone wants to estimate the probability that tomorrow's return will be less than c. Describe steps as to how a model could be constructed and the quantity of interest estimated.

(1) Obtain data on daily returns for this stock

(2) Choose a model for these data, e.g. Normal(μ, σ^2)

(3) Use data to estimate μ, σ^2, call these $\hat{\mu}, \hat{\sigma}^2$

(4) The prob. of interest is some function of μ, σ^2, i.e. $g(\mu, \sigma^2)$. So, estimate this as $g(\hat{\mu}, \hat{\sigma}^2)$

(5) Attach a measure of uncertainty to this estimate, i.e. stating a SE or a CI

Terminology

The parameters are the constants that distinguish probability models in the same "family." Examples of "families" include "normal," "exponential," and so forth.

A statistic is a function of the RVs being used to model the observable data. Examples include the sample mean, the sample variance, etc. A statistic is a **random variable**.

An estimator is a statistic which is used to estimate an unknown parameter or an unknown function of a parameter. It is also a **random variable**.

The "hat notation" is standard for denoting an estimator: $\hat{\beta}$ is an estimator for β, for example. θ is often used to denote a generic parameter, $\hat{\theta}$ is its estimator.

Exercise: Explain the sense in which an estimator is appropriately thought of as a random variable.

An estimator is a function of the RVs used to model the observable data.

Let $\hat{\theta}$ be ~~the~~ an estimator for θ.

$$\hat{\theta}(\omega) = g(X_1(\omega), X_2(\omega), \ldots, X_n(\omega)) \quad \omega \in \Omega$$

i.e. $\hat{\theta}$ is a function on Ω, it is a RV

can talk about $E(\hat{\theta})$, $V(\hat{\theta})$, $P(\hat{\theta} \leq c)$, ...

$$\bar{X}(\omega) = \frac{\sum X_i(\omega)}{n}$$, the sample mean

a natural estimator for the pop. mean

Performance of Estimators

Our objective is to estimate unknowns, i.e., to construct estimators. For any situation, there are many choices of estimator, so we require methods to assess the performance of an estimator.

The "performance" is closely tied to the distribution of the estimator. The distributions we learned in Probability (plus some more) are useful for this purpose.

An ideal estimator $\hat{\theta}$ for θ would have two properties:

1. The estimator should be accurate, i.e., its distribution would be centered on the true value θ.

2. The estimator should be precise, i.e., its distribution has a small spread (small variance)

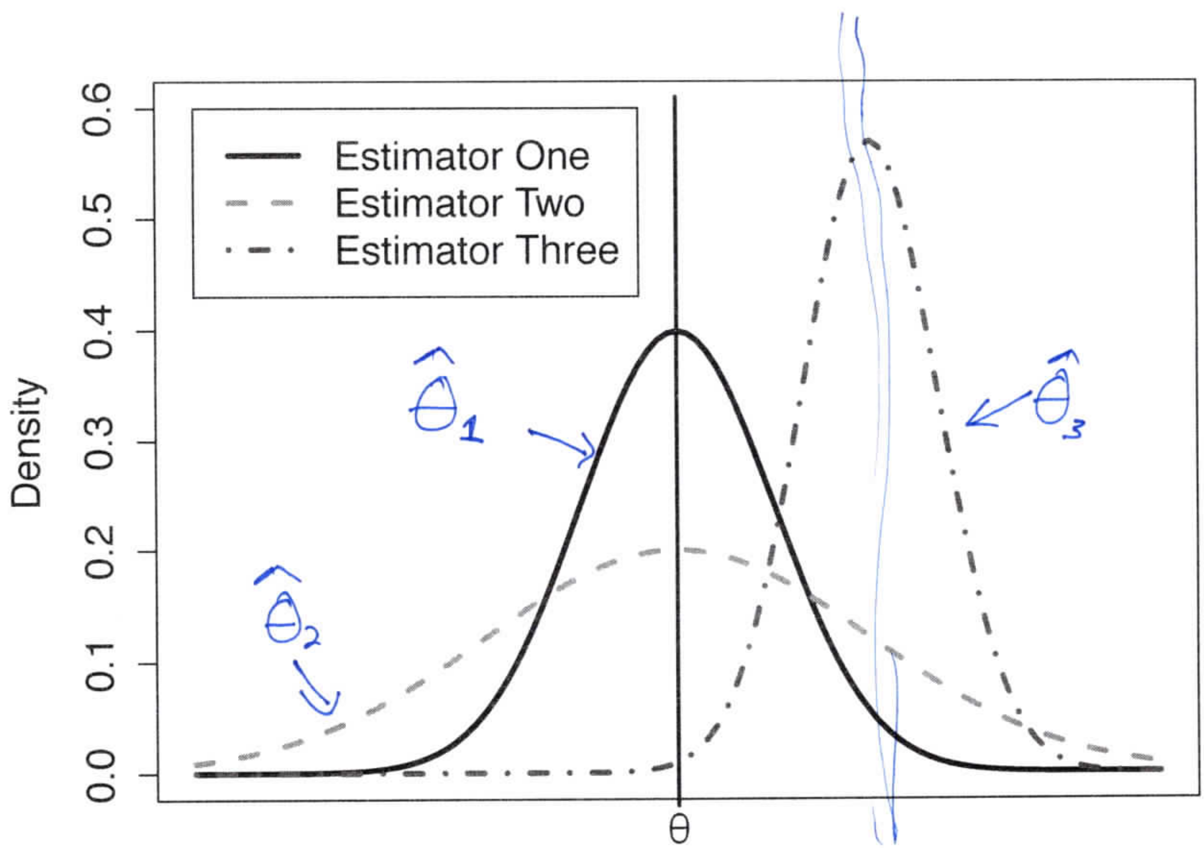
Example: The figure on the following page compares the distributions of three hypothetical estimators for a parameter θ, call them $\widehat{\theta}_1$, $\widehat{\theta}_2$, and $\widehat{\theta}_3$.

We note that $E(\widehat{\theta}_1) = \theta$, so it is optimally accurate. Also, $E(\widehat{\theta}_2) = \theta$.

But, $V(\widehat{\theta}_1) < V(\widehat{\theta}_2)$, so $\widehat{\theta}_1$ is more precise than $\widehat{\theta}_2$.

So, we would prefer $\widehat{\theta}_1$ to $\widehat{\theta}_2$.

Also, $V(\widehat{\theta}_3) < V(\widehat{\theta}_1)$, but $\widehat{\theta}_3$ has terrible accuracy.



An estimator $\widehat{\theta}$ is said to be an unbiased estimator for θ if $E(\widehat{\theta}) = \theta$.

The variability in an estimator (its precision) is naturally quantified by its variance, $V(\widehat{\theta})$.

The standard error (SE) of $\widehat{\theta}$ is the square root of its variance. The SE is often reported along with an estimate in order to give a sense of the amount of error one should anticipate in that estimate.

An estimator $\widehat{\theta}$ is said to be the minimum variance unbiased estimator for θ (MVUE) if $\widehat{\theta}$ is unbiased for θ and for any other unbiased estimator $\widehat{\theta}'$, $V(\widehat{\theta}) \leq V(\widehat{\theta}')$.

Exercise: Suppose that X has the binomial(n, p) distribution. It is common to estimate p using $\hat{p} = X/n$, the sample proportion. Is this estimator unbiased of $\hat{p}$? What is its variance and standard error?

$$E(\hat{p}) = E\left(\frac{X}{n}\right) = \frac{1}{n}E(X) = \frac{1}{n}(np) = p$$

Yes, $\hat{p}$ is an unbiased estimator of p ← "parameter"

"estimator"

$$V(\hat{p}) = V\left(\frac{X}{n}\right) = \frac{1}{n^2}V(X) = \frac{p(1-p)}{n}$$

$$\text{SE of } \hat{p} = \sqrt{V(\hat{p})} = \sqrt{\frac{p(1-p)}{n}}$$

often report approx. SE as $\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

There is often a "tradeoff" between bias and variance (accuracy and precision). This is called the bias-variance tradeoff.

In some cases, allowing some bias leads to reduced variance.

This tradeoff is made formal via the mean squared error (MSE). For an estimator $\widehat{\theta}$ for θ,

$$\mathrm{MSE}_\theta(\widehat{\theta}) = E\Big(\big(\widehat{\theta} - \theta\big)^2\Big)$$

i.e., it is the expected squared error in estimating θ when using $\widehat{\theta}$.

One can show that

$$\mathrm{MSE}_\theta(\widehat{\theta}) = V(\widehat{\theta}) + \underline{\mathrm{bias}_\theta^2(\widehat{\theta})}$$

where

$$\mathrm{bias}_\theta(\widehat{\theta}) = E(\widehat{\theta}) - \theta$$

$s^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$, $E(s^2) = \sigma^2$ under general conditions

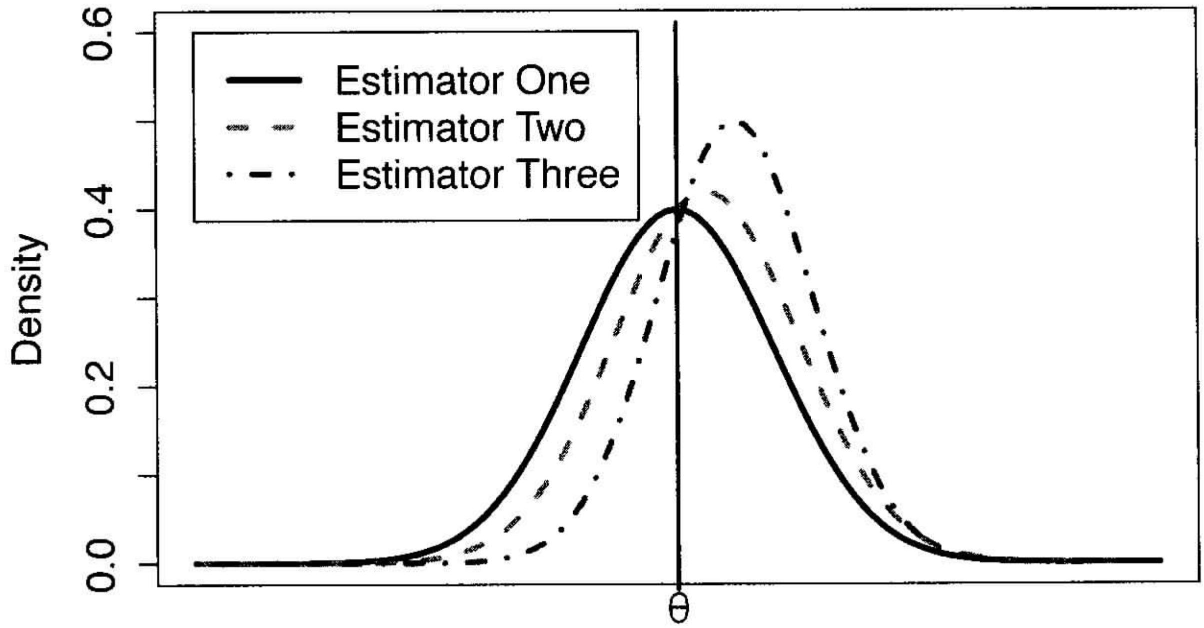
$\hat{\sigma}^2 = \frac{1}{n+1}\sum(X_i - \bar{X})^2$, minimizes MSE when $X_1, X_2, \dots X_n$ iid Normal

Exercise: How does the bias/variance tradeoff relate to our initial discussion of smoothness?

smooth models have lower variance, i.e. don't fluctuate as much as a new sample is drawn, but they have more bias, at greater risk of missing real features

Less smooth models have larger variance, as they try to fit to features/artifacts in each new sample drawn, but they have smaller bias

In the example below, all three estimators have the same MSE.



Important Cases

There are a few estimation situations that worth special attention.

Case 1: The first we have already explored, that of estimating a population proportion. We found that the sample proportion is unbiased and has a standard error of

$$\sqrt{\frac{p(1-p)}{n}}$$

Of course, we do not know the value of p, so typically it is replaced by its best estimate and the approximate SE is reported as

$$\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}$$

The Normal Distribution and the MGF

If X has the normal(μ, σ^2) distribution (with pdf given previously in the notes), then

$$M_X(t) = e^{t\mu + t^2\sigma^2/2}, \quad -\infty < t < \infty$$

The following properties of the normal(μ, σ^2) distribution are easy to prove once we know the MGF and its properties:

1. $E(X) = \mu$
2. $E(X^2) = \mu^2 + \sigma^2$, and hence $V(X) = \sigma^2$
3. If $Y = aX + b$, with $a \neq 0$, then $Y \sim N(a\mu + b, a^2\sigma^2)$.
4. The linear combination of independent normal random variables also has the normal distribution. In particular,
   (a) If $X_1, X_2, \ldots, X_n$ are i.i.d. $N(\mu, \sigma^2)$, then $\sum_i X_i \sim N(n\mu, n\sigma^2)$.
   (b) If $X_1, X_2, \ldots, X_n$ are i.i.d. $N(\mu, \sigma^2)$, then $\overline{X} \sim N(\mu, \sigma^2/n)$.
5. **The Central Limit Theorem:** Suppose $X_1, X_2, \ldots$ are i.i.d. and $E(X_i) = \mu$ and $V(X_i) = \sigma^2$ both exist and are finite, then

   $$\frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{D} Z$$

   where $\overline{X}_n = \sum_{i=1}^{n} X_i/n$ and $Z \sim N(0, 1)$.

From DeGroot and Schervish, Fourth Ed.

and

$$\sum_{i=1}^{99} p_i(1-p_i) = \frac{1}{100}\sum_{i=1}^{99} i - \frac{1}{(100)^2}\sum_{i=1}^{99} i^2$$
$$= 49.5 - \frac{1}{(100)^2}\cdot\frac{(99)(100)(199)}{6} = 16.665.$$

It follows from the central limit theorem that the distribution of the total number of questions that are answered correctly, which is $\sum_{i=1}^{99} X_i$, will be approximately the normal distribution with mean 49.5 and standard deviation $(16.665)^{1/2} = 4.08$. Therefore, the distribution of the variable

$$Z = \frac{\sum_{i=1}^{n} X_i - 49.5}{4.08}$$

will be approximately the standard normal distribution. It follows that

$$\Pr\left(\sum_{i=1}^{n} X_i \geq 60\right) = \Pr(Z \geq 2.5735) \simeq 1 - \Phi(2.5735) = 0.0050.$$ ◄

Outline of Proof of Central Limit Theorem

Convergence of the Moment Generating Functions Moment generating functions are important in the study of convergence in distribution because of the following theorem, the proof of which is too advanced to be presented here.

Theorem 6.3.5 Let $X_1, X_2, \ldots$ be a sequence of random variables. For $n = 1, 2, \ldots$, let F_n denote the c.d.f. of X_n, and let ψ_n denote the m.g.f. of X_n.

Also, let X^* denote another random variable with c.d.f. F^* and m.g.f. ψ^*. Suppose that the m.g.f.'s ψ_n and ψ^* exist ($n = 1, 2, \ldots$). If $\lim_{n\to\infty} \psi_n(t) = \psi^*(t)$ for all values of t in some interval around the point $t = 0$, then the sequence $X_1, X_2, \ldots$ converges in distribution to X^*. ■

In other words, the sequence of c.d.f.'s $F_1, F_2, \ldots$ must converge to the c.d.f. F^* if the corresponding sequence of m.g.f.'s $\psi_1, \psi_2, \ldots$ converges to the m.g.f. ψ^*.

Outline of the Proof of Theorem 5.7.1 We are now ready to outline a proof of Theorem 6.3.1, which is the central limit theorem of Lindeberg and Lévy. We shall assume that the variables $X_1, \ldots, X_n$ form a random sample of size n from a distribution with mean μ and variance σ^2. We shall also assume, for convenience, that the m.g.f. of this distribution exists, although the central limit theorem is true even without this assumption.

For $i = 1, \ldots, n$, let $Y_i = (X_i - \mu)/\sigma$. Then the random variables $Y_1, \ldots, Y_n$ are i.i.d., and each has mean 0 and variance 1. Furthermore, let

$$Z_n = \frac{n^{1/2}(\overline{X}_n - \mu)}{\sigma} = \frac{1}{n^{1/2}}\sum_{i=1}^{n} Y_i.$$

We shall show that Z_n converges in distribution to a random variable having the standard normal distribution, as indicated in Eq. (6.3.1), by showing that the m.g.f. of Z_n converges to the m.g.f. of the standard normal distribution.

If $\psi(t)$ denotes the m.g.f. of each random variable Y_i $(i = 1, \dots, n)$, then it follows from Theorem 4.4.4 that the m.g.f. of the sum $\sum_{i=1}^{n} Y_i$ will be $[\psi(t)]^n$. Also, it follows from Theorem 4.4.3 that the m.g.f. $\zeta_n(t)$ of Z_n will be

$$\zeta_n(t) = \left[\psi\left(\frac{t}{n^{1/2}}\right)\right]^n.$$

In this problem, $\psi'(0) = E(Y_i) = 0$ and $\psi''(0) = E(Y_i^2) = 1$. Therefore, the Taylor series expansion of $\psi(t)$ about the point $t = 0$ has the following form:

$$\begin{aligned}\psi(t) &= \psi(0) + t\psi'(0) + \frac{t^2}{2!}\psi''(0) + \frac{t^3}{3!}\psi'''(0) + \cdots \\ &= 1 + \frac{t^2}{2} + \frac{t^3}{3!}\psi'''(0) + \cdots.\end{aligned}$$

Also,

$$\zeta_n(t) = \left[1 + \frac{t^2}{2n} + \frac{t^3\psi'''(0)}{3!n^{3/2}} + \cdots\right]^n. \tag{6.3.15}$$

Apply Theorem 5.3.3 with $1 + a_n/n$ equal to the expression inside brackets in (6.3.15) and $c_n = n$. Since

$$\lim_{n\to\infty}\left[\frac{t^2}{2} + \frac{t^3\psi'''(0)}{3!n^{1/2}} + \cdots\right] = \frac{t^2}{2},$$

it follows that

$$\lim_{n\to\infty} \zeta_n(t) = \exp\left(\frac{1}{2}t^2\right). \tag{6.3.16}$$

Since the right side of Eq. (6.3.16) is the m.g.f. of the standard normal distribution, it follows from Theorem 6.3.5 that the asymptotic distribution of Z_n must be the standard normal distribution.

An outline of the proof of the central limit theorem of Liapounov can also be given by proceeding along similar lines, but we shall not consider this problem further here.

Summary

Two versions of the central limit theorem were given. They conclude that the distribution of the average of a large number of independent random variables is close to a normal distribution. One theorem requires that the random variables all have the same distribution with finite variance. The other theorem does not require that the random variables be identically distributed, but instead requires that their third moments exist and satisfy condition (6.3.9). The delta method lets us find the approximate distribution of a smooth function of a sample average.

Exercise: What is the practical value of the CLT?

When n is "large", $\overline{X}_n \stackrel{\sim}{\approx} N(\mu, \sigma^2/n)$

i.e. $\dfrac{\overline{X}_n - \mu}{\sigma/\sqrt{n}} \stackrel{\sim}{\approx} N(0,1)$

Exercise: Let $X_1, X_2, \ldots, X_n$ be i.i.d. random variables, each with the Exponential(1) distribution. What is the distribution of $\overline{X}$?

$E(X_i) = 1 = \mu$, $V(X_i) = 1 = \sigma^2$

$\sum_{i=1}^{n} X_i \sim \text{Gamma}(n, 1)$

$\overline{X} = \dfrac{\sum X_i}{n} \sim \text{Gamma}(n, n)$

The CLT says that

$\overline{X} \stackrel{\sim}{\approx} N(\mu, \frac{\sigma^2}{n}) = N(1, 1/n)$

for n "large"

Exercise: Suppose that on any day the price of a particular stock either increases by 2% (with probability 0.5) or decreases by 1% (with probability 0.5). Each day's change is independent of the other days. What is the probability that the stock's price is more than twice its current price 100 trading days from now?

Let $P_{100} = X_1 X_2 \cdots X_{100}$ where X_i are iid and $X_i = \begin{cases} 1.02 & \text{w.p. } 1/2 \\ 0.99 & \text{w.p. } 1/2 \end{cases}$

P_{100} = return after 100 days

$\{\text{stock price doubles after 100 days}\} = \{P_{100} \geq 2\}$

$$\log P_{100} = \sum_{i=1}^{100} \log X_i = \sum_{i=1}^{100} Y_i$$

where $Y_i = \begin{cases} \log(1.02) & \text{w.p. } 1/2 \\ \log(0.99) & \text{w.p. } 1/2 \end{cases}$

$Y_1, Y_2, \ldots, Y_{100}$ are i.i.d.

$\log(1+x) \approx x$ for x small

$\mu = E(Y_i) = 0.00488 = (\log(1.02))(\frac{1}{2}) + (\log(0.99))(\frac{1}{2})$

$\sigma^2 = V(Y_i) = 0.000223$

CLT says that $\bar{Y} \approx N(\mu, \sigma^2/100)$

Note that $P_{100} \geq 2$ iff $\log(P_{100}) \geq \log(2)$

iff $\sum Y_i \geq \log(2)$

iff $\bar{Y} \geq \log(2)/100$

So, $P(P_{100} > 2) = P(\bar{Y} > \log(2)/100)$

Exercise: A fair coin is tossed 1000 times. What is the probability that the number of heads is between 475 and 525?

$X = \#$ of heads $\sim$ binomial$(1000, 0.5)$

$X = \sum_{i=1}^{1000} U_i$ where $U_i = \begin{cases} 1, & \text{w.p. } 1/2 \\ 0, & \text{w.p. } 1/2 \end{cases}$

i.e. $U_1, U_2, \ldots, U_{1000}$ are iid

So, $\bar{X} \approx N\left(\overset{\mu}{0.5}, \frac{\overset{\sigma^2}{0.25}}{\underset{n}{1000}}\right)$ by CLT

$P(475 \le X \le 525) = P(0.475 \le \bar{X} \le 0.525)$

= use normal dist.

keep in mind: If toss coin n times, # of heads is approx. normal with mean $n/2$ and SD $\left(0.5/\sqrt{n}\right) n = \sqrt{n}\, 0.5$